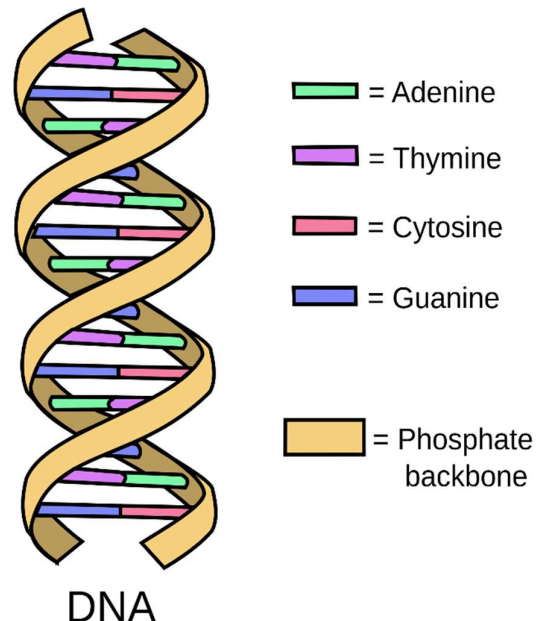


REPRODUCTION

- **Reproduction** is the biological process by which living organisms produce new individuals, or offspring, to ensure the continuation of their species.
- Reproduction involves the transfer of genetic material from parent organisms to their offspring
- Reproduction is not essential for the survival of an individual organism, unlike crucial life processes such as nutrition, respiration, or excretion.
- Reproduction requires considerable amount of energy from the body.
- Organism reproduce because of the survival of that species.
- To determine whether two different individual organisms belong to the same species, we typically observe their similarities.
- **Types of Reproduction:** Reproduction can be classified into two main types-
 - a) **Asexual Reproduction:** It is a type of reproduction that involves a single organism producing offspring without the involvement of another organism.
 - b) **Sexual Reproduction:** Sexual reproduction is a type of reproduction that involves the combination of genetic material from two parent organisms to produce offspring.
- **DNA (Deoxyribonucleic Acid):**
 - ✓ DNA is a molecule that carries the genetic instructions used in the growth, development, functioning, and reproduction of all living organisms.
 - ✓ DNA is also called as the blueprint or code for an organism's traits.
 - ✓ **Structure of DNA:** has a double-helix structure, resembling a twisted ladder. This structure consists of two long strands running in opposite directions, connected by pairs of nitrogenous bases.
 - ✓ **Location of DNA:** DNA is found in the nucleus of eukaryotic cells (or in the nucleoid region of prokaryotic cells).
 - ✓ **Components of DNA:** Each DNA strand is composed of a sugar-phosphate backbone and four types of nitrogenous bases:
 - ✓ Adenine (A)
 - ✓ Thymine (T)
 - ✓ Cytosine (C)
 - ✓ Guanine (G)
 - ✓ The bases pair specifically as Adenine pairs with Thymine, and Cytosine pairs with Guanine.
 - ✓ **Genetic Code:** The sequence of these nitrogenous bases encodes genetic information. This sequence determines the instructions for building proteins and influencing an organism's traits.
 - ✓ **Replication of DNA:** DNA can make copies of itself so that genetic information can be passed on to the next generation. When a cell divides, the DNA untwists and splits into two strands. Each strand then acts as a template to create a new, matching strand, resulting in two complete DNA molecules.
 - ✓ **Function of DNA:**
 - ✚ **Inheritance:** Passing genetic traits from parents to offspring.



- ✚ **Protein Synthesis:** Directing the production of proteins, which perform various functions within the cell.



➤ **DNA Copying and Variations:**

✓ **DNA Copying:**

- ✚ A fundamental event in reproduction is the copying of DNA.
- ✚ Cells use chemical reactions to replicate their DNA, resulting in two copies. These copies must be separated into new cells.

✓ **Variations in DNA:**

- ✚ Cells use chemical reactions to build copies of their DNA. Since no bio-chemical reaction is absolutely reliable, variations occur each time DNA is copied.
 - ✚ Variation is the differences found within the same species.
 - ✚ It can be caused by both sexual reproduction and environmental influences.
 - ✚ In some cases, DNA copies can have small differences from the original.
 - ✚ Some DNA copying variations can be so drastic that the new DNA copy cannot work with the inherited cellular apparatus, causing the newborn cell to die.
 - ✚ Many of variations are minor and don't stop the cells from working properly. Therefore, the new cells are similar to the original ones but have small variations.
 - ✚ These small variations during reproduction are important for evolution.
- ✓ Simply pushing one copy of DNA out of the original cell does not work because the pushed-out copy would lack an organized cellular apparatus to maintain life processes. Therefore, DNA copying is always accompanied by the creation of an additional cellular apparatus before the cell divides into two.
- ✓ **Genetic Variations contribution in survival of species:**
- ✚ Populations of organisms occupy specific places or niches within an ecosystem and rely on their ability to reproduce to maintain their presence in these niches. Niches can change due to factors beyond the organisms' control, such as fluctuations in temperature, changes in water levels, or meteorite impacts. If a niche changes drastically and a population is adapted to the old niche, that population might be wiped out. If some individuals in the population have variations that let them adapt to the new conditions, they might survive. So, having genetic variation is important for species to survive over time.
 - ✚ **Bacterial Adaptation:** If a population of bacteria lives in temperate waters and the water temperature rises due to global warming, most of the bacteria may die. However, some bacteria that can tolerate higher temperatures might survive and continue to grow.

➤ **Asexual Reproduction:**

- ✓ Asexual reproduction is a type of reproduction that involves only one parent organism. The offspring produced are genetically identical to the parent, meaning there are negligible genetic variation introduced during the process.
- ✓ Asexual reproduction is observed in both multicellular and unicellular organisms.
- ✓ **Characteristics of Asexual Reproduction:**
- ✚ **Single Parent:** Only one parent organism is involved in the reproduction process.
 - ✚ **No Fertilization:** There is no need for fertilization or the formation of gametes.
 - ✚ **Rapid Reproduction:** The process occurs quickly, allowing for fast reproduction.
 - ✚ **Rapid Growth:** Organisms multiply and grow at a rapid rate.
 - ✚ **Genetic Similarity:** The offspring produced are genetically identical to the parent organism.

✓ **Types of Asexual Reproduction:** Various types of Asexual reproduction are-

- (i) Fission
- (ii) Budding
- (iii) Fragmentation
- (iv) Regeneration
- (v) Spore Formation
- (vi) Vegetative Propagation

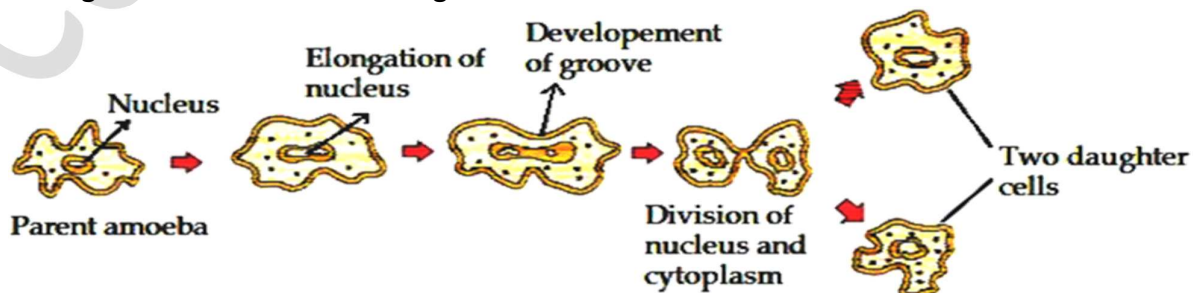
✓ **Fission:**

- ✚ Fission reproduction is a type of asexual reproduction in which a single organism divides into two or more new individuals.
- ✚ Fission is generally seen in unicellular organisms.
- ✚ Each new individual is identical of the original organism.
- ✚ Fission involves the splitting of the parent cell or body into two or more parts, each of which grows into a new, independent organism.

✚ **Types of Fission:**

⇒ **Binary Fission:**

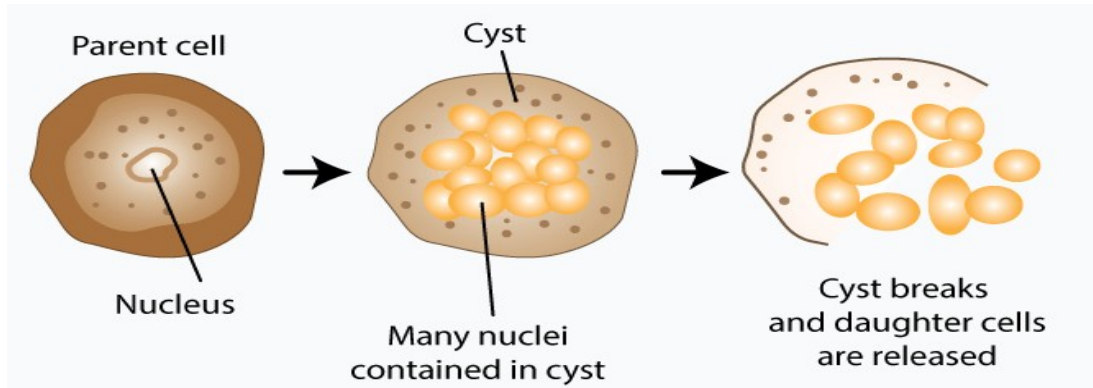
- It is a type of asexual reproduction in which a unicellular organism divides by splitting into two equal halves.
- This process results in two genetically identical offspring, both of which are almost clones of the original organism.
- Binary fission allows for rapid reproduction and population growth.
- Examples of Organisms which reproduces by Binary Fission:
 - ❑ **Bacteria:** Most bacteria reproduce by binary fission. For example, Escherichia coli (E. coli) divides in this manner. E. Coli found in intestine of organisms.
 - ❑ **Protozoa:** Single-celled protozoans like Amoeba and Paramecium reproduce through binary fission.
 - ❑ **Protozoa with specialized Structure:** *Leishmania* (which causes kala-azar), have specialized structures such as a whip-like flagellum at one end. In these organisms, binary fission occurs in a specific orientation relative to their body structure.
- **Steps in the process of Binary Fission (Reproduction of Amoeba):**
 - ❑ **DNA Replication:** The organism's DNA is replicated, so there are two copies of the genetic material.
 - ❑ **Cell Elongation:** The cell or body of the organism elongates.
 - ❑ **Division:** The cell or body divides into two separate parts. In unicellular organisms, this often involves the splitting of the cell membrane or cell wall.
 - ❑ **Formation of Two New Individuals:** Each part develops into a complete, independent organism with its own set of genetic material.



⇒ **Multiple Fission:**

- Multiple fission is a form of asexual reproduction in which a single organism divides into many daughter cells simultaneously.

- In multiple fission, a cyst is a protective shell that surrounds an organism. It helps the organism survive tough conditions and allows it to produce many offspring inside.
- Multiple fission results in the formation of several genetically identical offspring from one parent.
- Multiple fission allows for rapid reproduction and population growth.
- Examples of Organisms which reproduces by Multiple Fission:
- *Plasmodium*, which causes malaria, undergoes multiple fission.



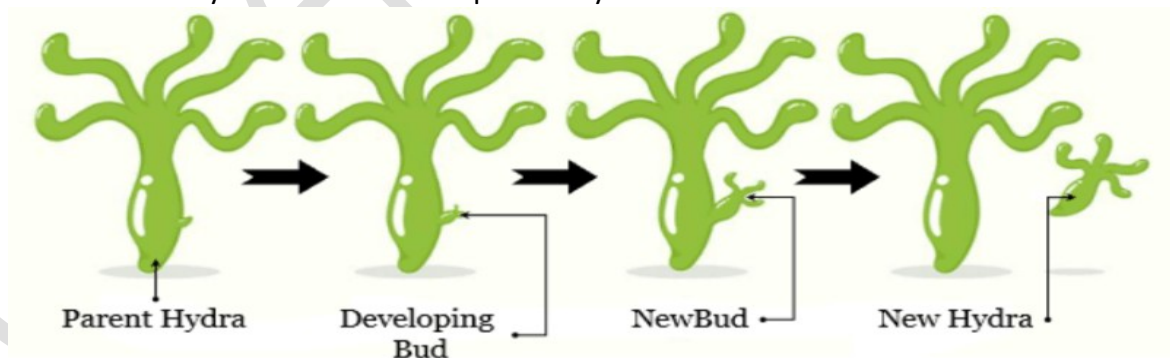
✓ Budding:

- ✚ Budding is a form of asexual reproduction where a new organism develops as an outgrowth, or "bud," from the parent organism.
- ✚ Budding uses special cells called regenerative cells to reproduce.
- ✚ Budding can take place in both unicellular and multicellular organisms.
- ✚ Process of budding:

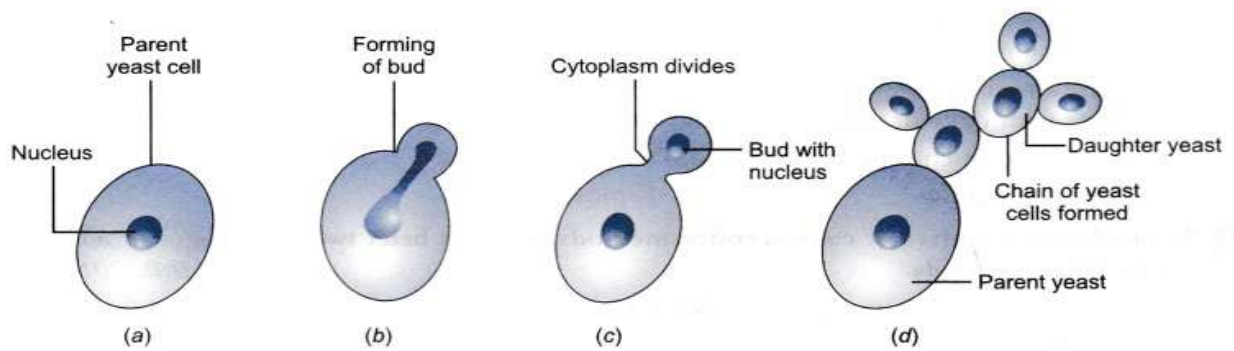
- ❓ **Growth:** A small part of the parent organism grows outward due to repeated cell divisions at a specific spot., forming a "bud."
- ❓ **Development:** This bud slowly grows into a new, complete organism.
- ❓ **Separation:** When the new organism is fully developed, it breaks away from the parent.
- ❓ **New Organism:** The separated bud becomes a new, independent organism.

✚ Example:

- ❓ **Hydra:** Small outgrowths, or buds, form on the parent Hydra. These buds grow into new Hydras and eventually detach to live independently.

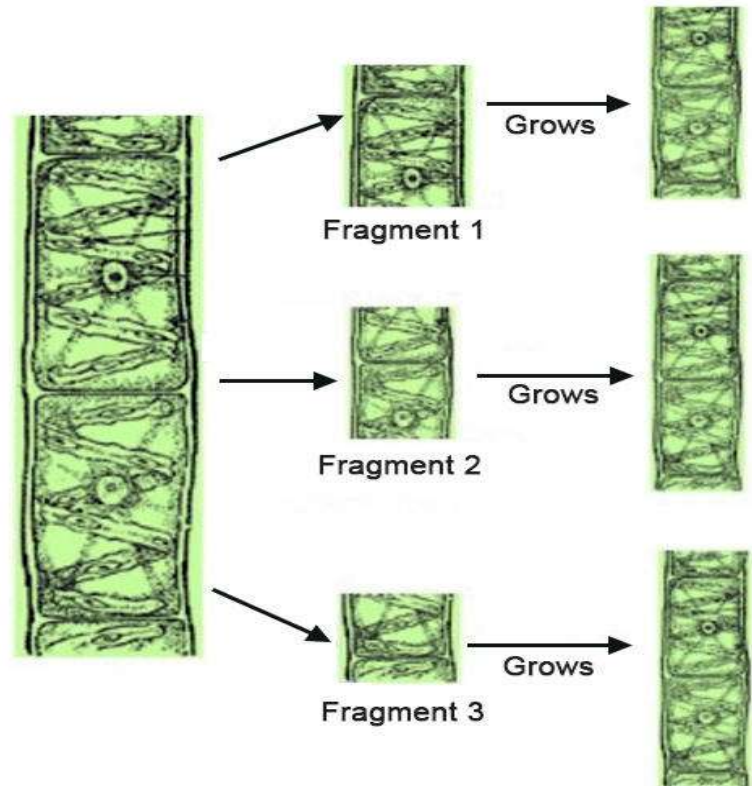


- ❓ **Yeast:** In a warm place, yeast cells rapidly reproduce in a 10% sugar solution by forming small outgrowth buds that separate and grow into new independent yeast cells.



✓ **Fragmentation:**

- ✚ Fragmentation is a type of asexual reproduction in which an organism breaks into several pieces (fragments), with each piece capable of growing into a new individual.
- ✚ Each fragment contains part of the original organism's tissues and, in some cases, essential organs.
- ✚ Complex organisms with specialized cells and tissues cannot use basic cell-by-cell division for reproduction. Reproduction requires specific cells that can grow, divide and differentiate into all necessary cell types to form a new organism. These cells are Reproductive cells.
- ✚ It can be seen in multicellular organisms.
- ✚ It allows for rapid population growth under favourable conditions.
- ✚ The new organisms are genetically identical to the parent.
- ✚ Examples of Organisms which reproduce by Fragmentation:



Spirogyra – It is filamentous structured algae found in dirty pond water that reproduces by breaking into smaller pieces upon maturation. Each fragment then grows into a new individual.

✓ **Regeneration:**

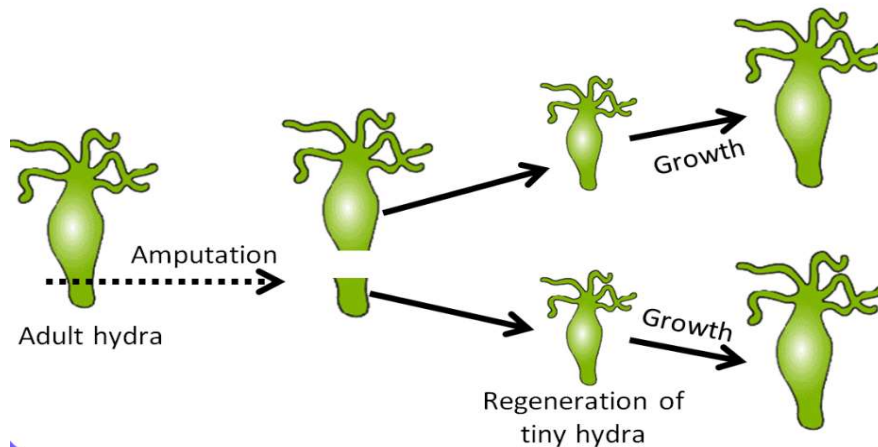
- ✚ Regeneration is the biological process by which an organism can grow back lost or damaged body parts, or produce new individuals from fragments of its body.
- ✚ Many fully differentiated organisms can give rise to new individuals from their body parts. If these organisms are cut or broken into pieces, many of these pieces can develop into separate individuals.
- ✚ Removal of a body part is called Amputation.
- ✚ Regeneration Process:
 - ② **Specialized Cells:** Regeneration involves specialized cells that are capable of proliferation (process by which cells undergo repeated cycles of division to produce new cells).
 - ② **Cell Proliferation:** These specialized cells multiply to form a large number of new cells.

❓ **Development:** From this mass of new cells, different cells undergo changes to form various cell types and tissues in a well-organized sequence.

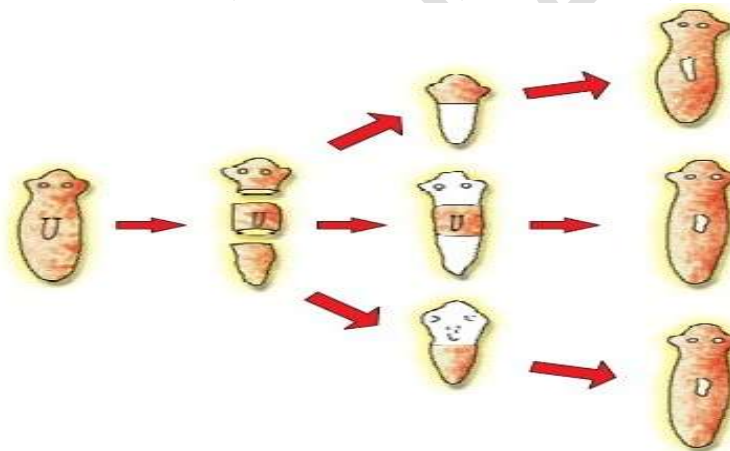
✚ Regeneration is different from reproduction. While regeneration can occur when an organism is cut into pieces, most organisms do not rely on being cut up for reproduction.

✚ **Examples:**

❓ **Hydra:** This simple animal can be cut into several pieces, and each piece can grow into a completely new organism. Normally Hydra can reproduce by budding.



❓ **Planaria:** Similar to Hydra, Planaria can be fragmented, and each piece will regenerate into a whole new Planaria. Normally Planarias can reproduce sexually.



✓ **Spore Formation (Sporogenesis):**

✚ Spore Formation is a process that produces reproductive cells (spores) that can grow into new organisms.

✚ **Process of Spore Formation:**

❓ Many Spores are housed in sacs known as Sporangia.

❓ The plants produce hundreds of spores and the spore sac bursts.

❓ These spores are dispersed into the air, where they germinate and create a new plant under favourable conditions.

✚ **Rhizopus (bread mould)** grows best on a wet slice of bread kept in a cool, moist, and dark place.

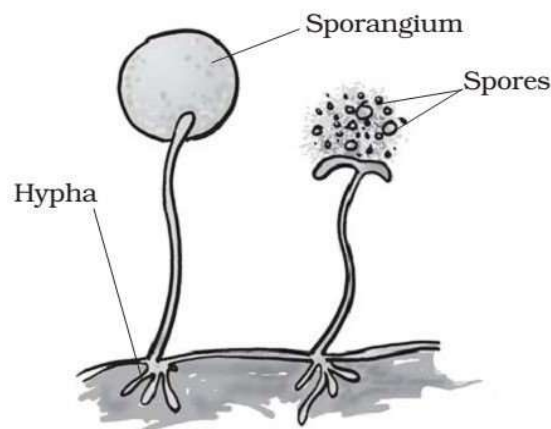
✚ Structure of Rhizopus:

⇒ Non-Reproductive Structures:

- ❓ **Hyphae:** The thread-like structures seen on the bread are called hyphae. They are part of the bread mould (Rhizopus) but are not involved in reproduction.

⇒ Reproductive Structures:

- ❓ **Sporangia:** The small sac like blob-on-a-stick structures are called Sporangia. They are involved in reproduction.
- ❓ **Spores:** Tiny cells inside the sporangia called spores. Spores can develop into new Rhizopus moulds.



⇒ Spores protective walls:

- ❓ **Thick Walls:** Spores are covered by thick walls that protect them until they find a moist surface to grow on.

✓ Vegetative Propagation:

- ✚ On the basis of reproduction, plant parts are divided into two categories:

- Vegetative parts:** The vegetative parts of a plant are the parts that do not participate in sexual reproduction, and include the roots, stems, and leaves.
- Reproductive parts:** The reproductive parts of a plant are the parts that participate in sexual reproduction, and include the flower, which contains the stamen (male part) and pistil (female part).

- ✚ Vegetative propagation is a type of asexual reproduction in plants where a new plant grows from a fragment of the parent plant.
- ✚ It can occur through the regeneration and fragmentation of specific vegetative parts of the plant, such as the roots, stems, or leaves.
- ✚ All parts in every plant cannot develop into new individual but many vegetative parts of some plants can develop into new organism when given the right conditions required for the growth.

✚ Advantages of Vegetative Propagation:

- ❓ **Faster Growth and Development:** Plants raised through vegetative propagation often produce flowers and fruits sooner than those grown from seeds.
- ❓ **Seedless Plant Propagation:** These techniques also facilitate the propagation of plants that have lost the ability to produce seeds, including bananas, oranges, roses, and jasmines.
- ❓ **Genetic Consistency:** An advantage of vegetative propagation is that all new plants are genetically similar to the parent plant, retaining its characteristics.

✚ Types of Vegetative Propagation: There are two types of Vegetative propagation-

- Natural Vegetative Propagation
- Artificial Vegetative propagation

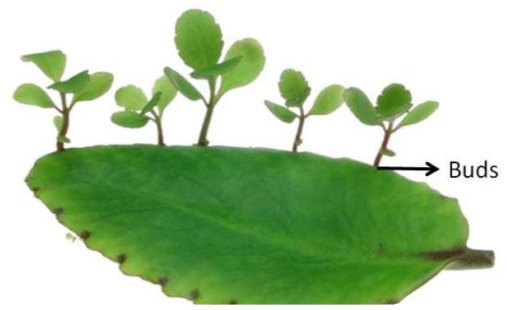
✚ Natural Vegetative Reproduction:

- ⇒ Natural vegetative propagation is a process by which plants grow and develop without human intervention.
- ⇒ It occurs when an axillary bud grows into a lateral shoot and develops its own roots.
- ⇒ The extra part that grows is called as adventitious part.

⇒ Modes of Natural Vegetative propagation:

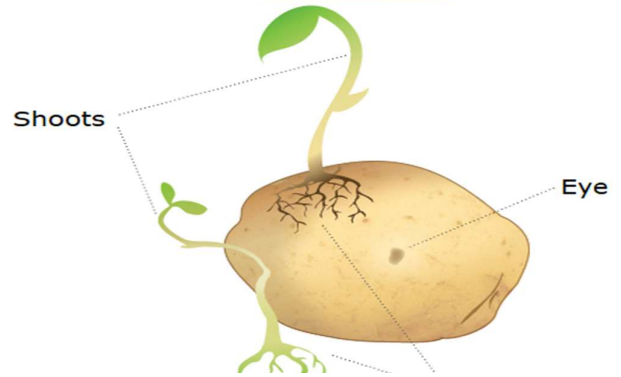
- **By Leaves:**

- ❑ In some plants vegetative propagation occurs through the leaves.
- ❑ Here buds grow on the margin of the leaves.
- ❑ These reproductive buds, when falling to the ground, germinate and form new plants.
- ❑ Examples are Bryophyllum leaf and Begonia Leaf etc.



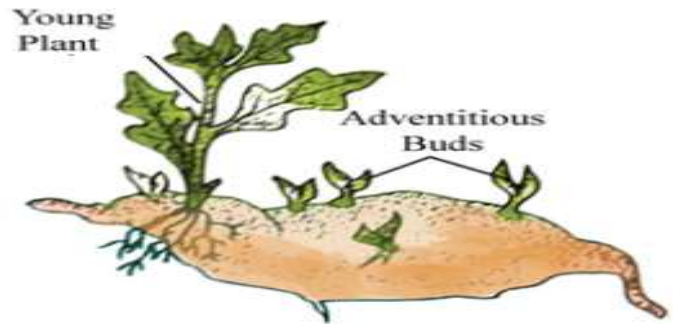
- **By Stem:**

- ❑ In some plants vegetative propagation occurs through the Stems.
- ❑ Here shoots grow on the parts of stem.
- ❑ These shoots fall to the ground, germinate and form new plants.
- ❑ Examples are Potato, Ginger, onion, Grass etc.



- **By Roots:**

- ❑ In some plants vegetative propagation occurs through the Roots.
- ❑ Here buds grow on the parts of Roots
- ❑ These buds fall onto the ground and germinate and form new plants.
- ❑ Examples are Sweet Potato, carrot etc.



✚ **Artificial Vegetative Reproduction:**

⇒ Artificial vegetative propagation is a method of plant reproduction that involves human intervention to create new plants from parts of existing plants.

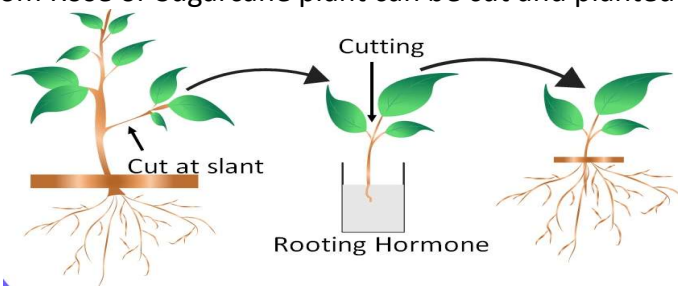
⇒ This is helpful when there are not many materials available for growing new plants.

⇒ Types of Artificial Vegetative Reproduction:

- (i) Cutting (ii) Grafting (iii) Layering (iv) Tissue Culture

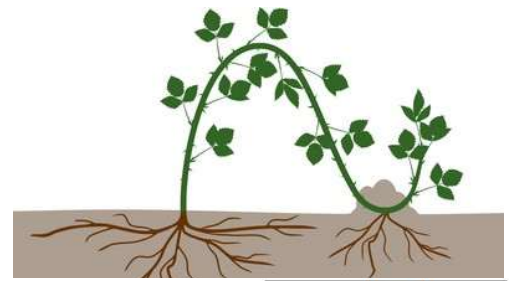
⇒ **Cutting:**

- ❑ A healthy young branch with a few nodes and internodes is cut and most of its leaves are removed.
- ❑ The cut end is stuck into moist soil, and the cutting will develop roots and grow into a new plant.
- ❑ A stem or leaf from Rose or Sugarcane plant can be cut and planted to grow new plant.



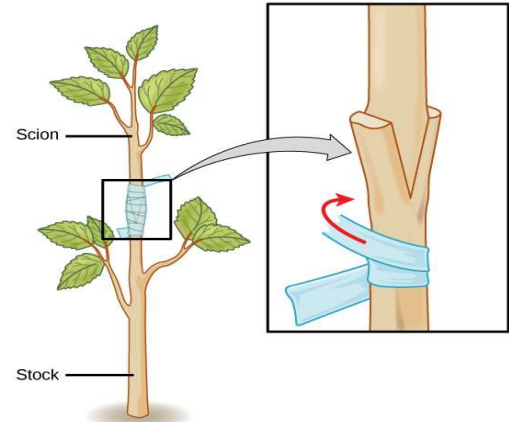
⇒ **Layering:**

- ❑ Layering is a plant propagation technique that involves encouraging new roots to form on a stem while it's still attached to the parent plant.
- ❑ Sometimes it occurs naturally in some plants.
- ❑ Examples: It can be done with jasmine, mint and Rose plant.



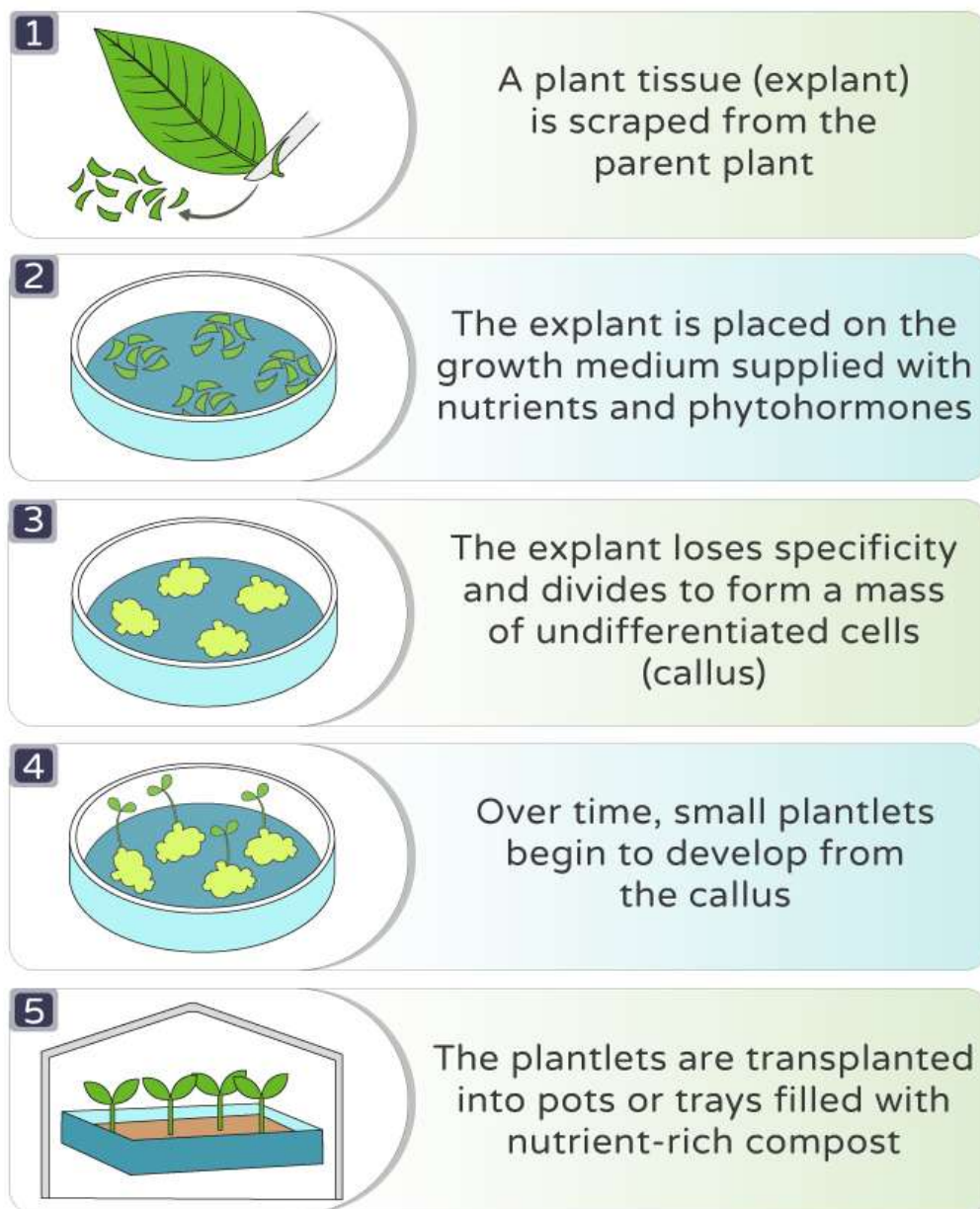
⇒ **Grafting:**

- ❑ Grafting is an artificial method of vegetative propagation that involves combining two plants to create a new variety.
- ❑ It helps in giving more than one desirable character in plant.
- ❑ A stem, bud, or branch from one plant, called the scion, is attached to the stem and roots of another plant, called the stock.
- ❑ The vascular tissue of the two plants combines, allowing the scion to grow.
- ❑ Plants of same family can be graft together.
- ❑ Example: Lemon and Orange can be graft together. Also, Mango varieties can be grafted together.



⇒ **Tissue Culture:**

- ❑ Tissue culture is a technique that involves growing cells, tissues, or organs in an artificial environment, separate from the original organism.
- ❑ It's also known as micropropagation.
- ❑ Tissue culture can be used for a plant propagation and research.
- ❑ Examples: Oil palm, banana, pineapple, tomato, sweet potato have been produced by tissue culture in the developing countries.
- ❑ The process of Tissue culture is explained in the figure:



➤ Sexual Reproduction

- ✓ Sexual reproduction is a biological process where two parent organisms create a new unique offspring.
- ✓ Sexual reproduction requires the involvement of two parents, typically one male and one female
- ✓ Each parent produces specialized reproductive cells called gametes (sperm in males and eggs in females).

- ✓ The male gamete (sperm) and the female gamete (egg) fuses during fertilization to form a new cell called a zygote.
- ✓ The zygote develops into a new individual through a series of growth and development stages.
- ✓ **DNA Copying and Variations in Reproduction:**
 - ✚ **DNA Copying Process:** When one cell divides into two new cells, it involves copying DNA and other cell components. This process isn't perfectly accurate, leading to variations.
 - ✚ **Importance of Variations:**
 - ⇒ Variations caused by copying errors help populations adapt and survive. Though individual organisms might not benefit directly, variations are crucial for the species as a whole.
 - ⇒ Variations build up over generations. When DNA from different individuals is combined during sexual reproduction, it creates new and unique combinations.
 - ✚ **Challenge of DNA Duplication:** If each generation combined DNA from two individuals, the amount of DNA would double each time. This would eventually disrupt cellular function and lead to too much DNA.
 - ✚ **Solution to DNA Duplication:** Many multicellular organisms have special cells called germ-cells that contain only half the DNA of regular body cells. When these germ-cells combine during reproduction, they restore the proper amount of DNA in the new individual.
 - ✚ **Germ-Cell:** In simple organisms, germ-cells may be similar. In more complex organisms, germ-cells specialize: one type stores food (female gamete), and the other is motile (male gamete).
 - ✚ **Impact on Reproductive Organs:** The differences in germ-cells lead to differences in male and female reproductive organs and in some species, differences between male and female bodies also.
- ✓ **Advantages of Sexual Reproduction:**
 - ✚ Sexual reproduction promotes genetic variation, which can improve a population's ability to adapt to changing environments
 - ✚ It resists diseases.
- ✓ **Disadvantages of Sexual Reproduction:**
 - ✚ It requires more energy as compared to asexual reproduction.
 - ✚ It requires more time as compared to asexual reproduction.
 - ✚ It has a limitation of finding a mate or partner.
- ✓ **Sexual Reproduction in plants**
 - ✚ Angiosperms and gymnosperms are two main groups of vascular seed plants that differ in how their seeds develop.
 - ✚ **Gymnosperms:**
 - ⇒ Gymnosperms are plants which produces seeds without fruit.
 - ⇒ They are non-flowering plants.
 - ⇒ Gymnosperms have reproductive structures known as cones.
 - ✚ **Angiosperms:**
 - ⇒ Angiosperms are plants that produce seeds enclosed within a fruit.
 - ⇒ They are flowering plants.
 - ⇒ Flowers are their reproductive structures.
 - ✚ **Types of Plant reproductive parts/Flower:**
 - **Unisexual Flowers:**
 - ❓ Unisexual flowers contain only one type of reproductive organ: either stamens (male parts) or carpels (female parts).
 - ❓ They do not have both stamens and carpels in the same flower.
 - ❓ Examples are Papaya and Watermelon.

■ **Bisexual Flowers:**

- ❑ Bisexual flowers contain both stamens (male parts) and carpels (female parts) within the same flower.
- ❑ They have both reproductive organs necessary for fertilization in a single structure.
- ❑ Examples are Hibiscus and Mustard.

🌺 **Reproductive Parts of Angiosperms:**

⇒ **Location of Reproductive Parts**

- The reproductive parts of angiosperms are found in the flower.
- The key reproductive structures are stamens (Male Parts) and carpels (Female Parts), which contain the germ-cells.

⇒ **Parts of Flower:**

○ **Petals:**

- Petals are colourful part of a flower.
- Functions of Petals:
 - ❑ Attracting insects and birds: Bright colours and patterns draw insects and birds to the flower.
 - ❑ Protect Reproductive Parts: Shields the stamens and carpels, especially before flower opens.

○ **Sepals:**

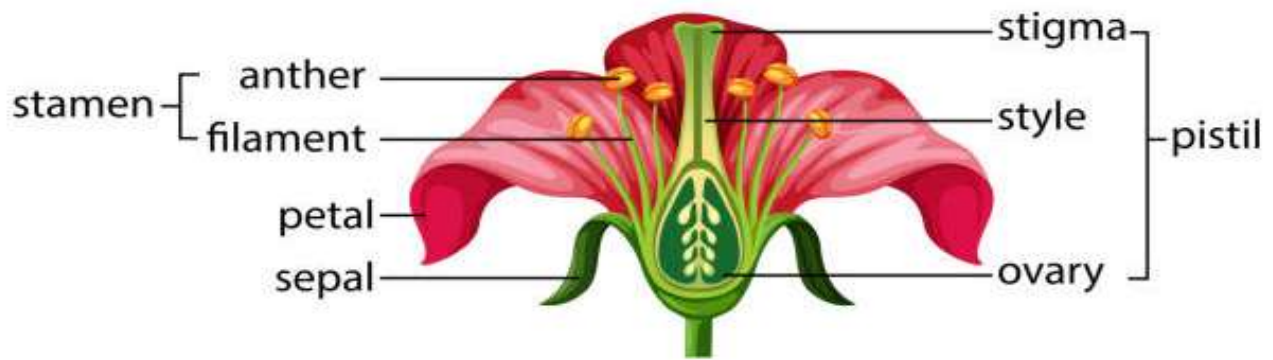
- Sepals are green, leaf-like structures beneath the petals.
- Functions of Sepals:
 - ❑ Protect the Bud: Shields the developing flower from damage and environmental factors.
 - ❑ Support the Flower: Provides structural support once the flower blooms.

○ **Male Reproductive Part: Stamen:**

- The stamen is the male reproductive organ in a flower.
- It consists of two main components: the **anther** and the **filament**.
 - (i) Anther:**
 - The anther is the upperpart of the stamen where yellowish powdery substance, pollen grains are produced.
 - Functions of Anther:
 - ❑ Pollen Production: The anther contains pollen sacs (microsporangia) where pollen grains are formed. Pollen grains which carry the male germ-cells. These grains carry the male gametes (sperm cells) necessary for fertilization.
 - ❑ Pollen Release: Once mature, the anther releases pollen into the environment, facilitating the transfer of male gametes to the female parts of the flower.

(ii) Filament:

- The filament is the stalk that supports the anther and connects it to the flower.
- Functions of Filament:
 - ❑ Supports the Anther: The filament lifts the anther so it can easily release pollen.
 - ❑ Nutrient Supply: It helps transport nutrients to the anther to help it produce pollen.



?

○ **Female Reproductive Part: Carpel (Pistil):**

- The carpel, also known as the pistil, is the female reproductive organ in a flower.
- It consists of three main parts: the ovary, style, and stigma.

(i) Ovary:

- The ovary is the swollen, basal portion of the carpel.
- It contains ovules, which are the female gametes (egg cells).
- Functions of Ovary:
 - ? Holds Egg Cells: Contains ovules (egg cells) that will be fertilized.
 - ? Fertilization: Receives pollen and helps fertilize the eggs, turning them into seeds.
 - ? Fruit Formation: After fertilization, it turns into a fruit, which protects and helps spread the seeds.
 - ? Nutrient Support: Provides nutrients to the developing seeds.

(ii) Style:

- The style is the elongated, tube-like structure that connects the ovary to the stigma.
- Functions of Style:
 - ? Pollen Passage: Allows pollen to travel from the stigma down to the ovary.
 - ? Supports Pollination: Provides a path for pollen tubes to reach the ovules for fertilization.
 - ? Nutrient Supply: Transports nutrients from the plant to the ovary to support seed development.

■ Stigma:

- The stigma is the sticky tip of the flower's carpel.
- Functions of Stigma
 - ? Catches Pollen: The sticky surface grabs pollen grains from pollinators or the wind.
 - ? Pollen Germination: Helps pollen start growing so it can move down to the ovary.
 - ? Starts Fertilization: Allows pollen to begin the process of fertilizing the egg cells in the ovary.

✚ **Various Process in Plant Reproduction:** It has various process:

1) Pollination:

- Pollination is the transfer of pollen from the male part of a flower (stamen) to the female part (carpel) of the same flower or another flower.
- It is necessary for fertilization to occur.
- It is a critical step in the reproductive process of flowering plants (angiosperms).

• Pollinators:

- ? Pollinators are organisms that facilitate the transfer of pollen from the male parts (stamens) of flowers to the female parts (stigmas) of the same or other flowers.

- ❑ Pollination by insects (e.g., bees, butterflies, beetles) which visit flowers for nectar and unintentionally transfer pollen.
- ❑ Pollination by birds (e.g., hummingbirds) that are attracted to brightly colored flowers and transfer pollen as they feed.
- ❑ Pollination by wind, where lightweight pollen is carried by air currents to other flowers.
- ❑ Pollination by water, where pollen is carried by water currents to other flowers, common in aquatic plants.

● Types of Pollination:

⇒ Self-Pollination:

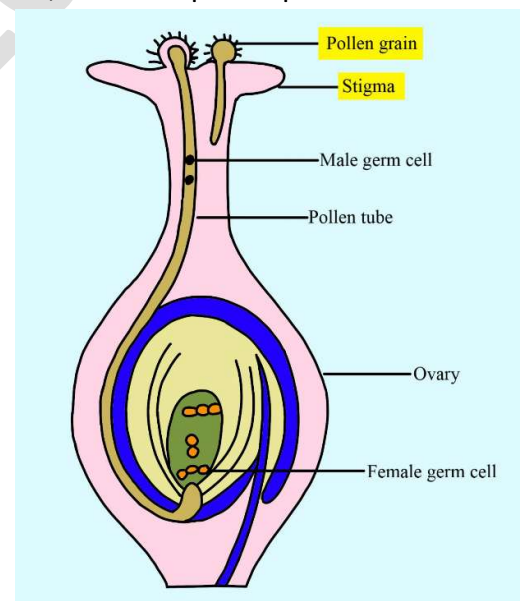
- ❑ It is a type of pollination in which pollen from the stamens of a flower lands on the stigma of the same flower or another flower on the same plant.
- ❑ Pollen transfers can occur within the same flower.
- ❑ Pollen transfers can occur between different flowers on the same plant.
- ❑ It guarantees fertilization if pollinators are available or not available.
- ❑ Examples are Tomatoes, peas etc.

⇒ Cross-Pollination:

- ❑ It is a type of pollination in which pollen is transferred from the stamens of one flower to the stigma of a different flower, often from a different plant of the same species.
- ❑ Pollen transfers can occur within the flowers of different plants.
- ❑ It Promotes genetic diversity, which can lead to healthier, more adaptable plants.
- ❑ Examples are Apples, cherries, sunflowers etc.

● Pollen Tube Formation/ Germination of Pollen on stigma:

- ❑ Pollen tube formation is the process where a pollen grain, after landing on a compatible stigma, grows a tube through the style to deliver sperm cells to the ovule.
- ❑ Multiple pollen grains can land and germinate on the stigma, with each producing a pollen tube that travels through the style toward the ovary..
- ❑ Pollen tube carries two sperm cells.
- ❑ Inside the ovule, the pollen tube releases two sperm cells into the embryo sac.



2) Fertilization Process:

- Fertilization in plants is the process by which male and female gametes (sperm cells and egg cells) fuse to form a zygote.
- In plants, two distinct fertilization events occurring simultaneously within the ovule of flowering plants, thus it is called as Double fertilisation.

● Double Fertilization:

⇒ Syngamy:

- ❑ One sperm cell from the pollen grain fuses with the egg cell (ovum) located in the embryo sac of the ovule.
- ❑ These fusion forms a zygote.
- ❑ Zygote is a diploid cell (Two sets of chromosomes).
- ❑ Zygote will develop into the embryo.

⇒ **Triple Fusion:**

- ❑ Sperm cell fuses with two polar nuclei located in the central cell of the embryo sac.
- ❑ This fusion results in a triploid cell (three sets of chromosomes).
- ❑ The triploid cell develops into the endosperm.
- ❑ Endosperm is a nutrient-rich tissue that supports the developing embryo within the seed.

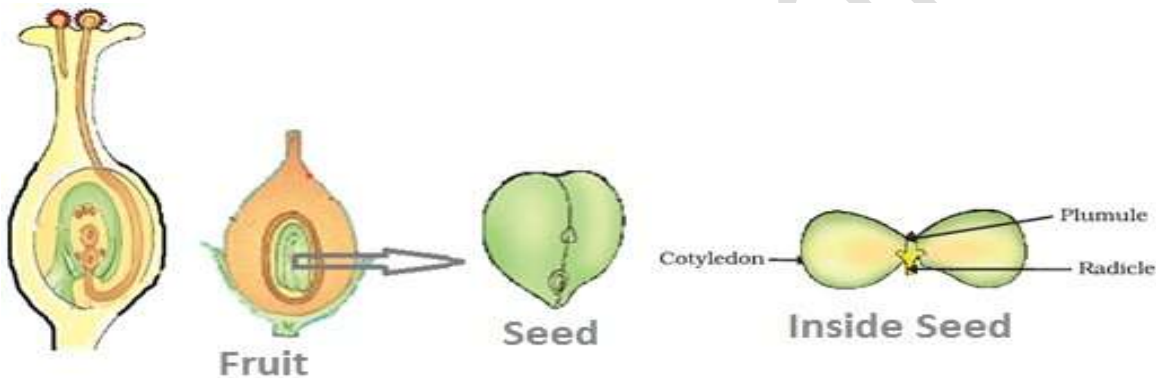
3) Seed Formation and Fruit Development:

- Post-fertilization, the zygote divides to form an embryo within the ovule.
- The ovule develops a tough coat and transforms into a seed.
- The ovary grows and ripens into a fruit.
- Other flower parts like petals, sepals, stamens, style, and stigma may shrivel and fall off.

4) Germination:

- Germination is the process by which a seed develops into a new plant.
- It involves the transition from a dormant state to active growth, leading to the emergence of the seedling (young plant).

● **Key Steps of Germination:**



- ❑ The seed takes in water, causing it to swell and kick-start its growth.
- ❑ **Root Development:** The radicle (the first part of the seedling) grows down into the soil to form the root, which helps the plant take up water and nutrients.
- ❑ **Shoot Development:** The plumule (the part that will become the stem and leaves) pushes up through the soil to start forming the plant's above-ground parts.
- ❑ **Cotyledons** are the first leaves that appear in a seedling and play a crucial role in the early stages of plant growth. They store nutrients that help the young plant grow until it can produce its own food.
- On the basis of Cotyledons, seeds are divided into two parts:
 - ❑ **Monocots (Monocotyledons):**
 - Number of Cotyledons: One cotyledon
 - Leaf Veins: Parallel veins
 - Examples: Grass, lilies, and orchids
 - ❑ **Dicots (Dicotyledons):**
 - Number of Cotyledons: Two cotyledons
 - Leaf Veins: Net-like or branching veins
 - Examples: Beans, sunflowers, and roses

✓ Sexual Reproduction in Human Beings

- Humans and many mammals reproduce by the mode of sexual reproduction.
- Primary reproductive organ in Males is Testis gland that produces Testosterone Hormone and male gamete(sperm).
- Primary reproductive organ in Females is Ovary which produces Oestrogen and Progesterone Hormones and female Gamete (Ovum).
- In sexual reproduction, germ-cells from two individuals of different gender need to meet and combine. This can happen through internal or external transfer, depending on the species.
- Stages in the life of a human:
 - (i) **Prenatal development:** The period of 280 days before birth
 - (ii) **Infancy:** From birth to 2 years of age
 - (iii) **Toddlerhood:** From 1-5 to five years of age
 - (iv) **Childhood:** From 3-11 years of age
 - (v) **Adolescence:** From 12 to 18 years of age
 - (vi) **Adulthood:** From 19 to 60 years of age
 - (vii) **Late Adulthood:** 60 years and older
 - (viii) **Dying and death:** The final stage of human life

• **Puberty:**

- Puberty is the period when the body matures sexually and prepares for the reproductive process.
- It is the time of life when child experiences physical and hormonal changes that leads to sexual maturity.
- Occurs when body growth slows down, between the ages of 10-13 in females and 11-14 in males.
- Initially, the body focuses on general growth, becoming larger. But as growth slows down, the reproductive organs mature.
- It is also termed as **Sexual Maturation** and it involves development of reproductive tissues.

• **Adolescence:**

- Adolescence is the period of life between childhood and adulthood, when a human goes through physical, sexual and psychological developments.
- It is usually associated with the teenage years, but it can start earlier or end later.

• **Changes During Adolescence:**

- ⇒ Body undergoes several changes during adolescence.
- ⇒ Changes occur over months/years, not simultaneously.
- ⇒ Each individual experiences growth differently.

⇒ **Common Changes:**

- ☐ Hair growth in armpits, genital areas, and limbs.
- ☐ Skin may become oily, leading to pimples.
- ☐ Increased height and body mass.
- ☐ Increased body awareness.
- ☐ New physical sensations and emotional responses.

⇒ **Changes in Male:**

- ☐ Development of thicker hair on the face (Mustache, beard).
- ☐ Voice deepens, often becomes crackly, Formation of Adam's Apple.
- ☐ Increased hair growth on the chest, abdomen, and legs.

- ☐ Enlargement of the penis and increased frequency of erections.
- ☐ Increase in muscle mass and strength.

⇒ **Changes in Female:**

- ☐ Onset of the menstrual cycle.
- ☐ Increased hair growth in armpits and the genital area.
- ☐ Enlargement of breasts and darkening of the skin around the nipples.
- ☐ Widening of hips and thighs and changes in body proportions.

● **Rate of Changes during Adolescence:**

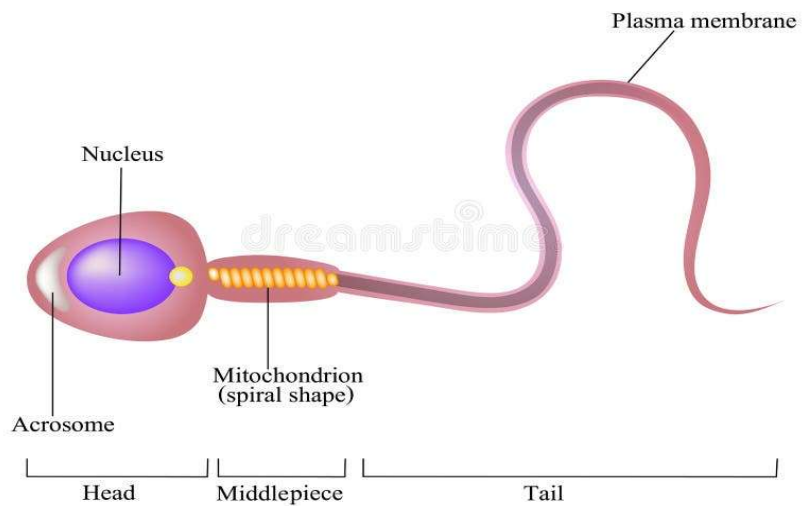
- ⇒ These changes occur gradually over several months or years. They do not happen at the same time or in the same way for every person.
- ⇒ Some individuals may experience these changes early and quickly, while others may experience them more slowly.
- ⇒ Girls usually grow faster between ages 10-14 and stop growing by 16-18. Boys grow faster a bit later, between ages 12-16, and keep growing until 18-21. Because of this, boys tend to be taller than girls.

🌈 **Male Reproductive System:**

- The male reproductive system includes a group of organs that make up the reproductive system and few parts of urinary system in people assigned male at birth.
- **Functions of Male reproductive system:**
 - ☐ Produce, maintain and transport sperm cells and semen. Sperm cells are male reproductive cells. Semen is the protective fluid around sperm.
 - ☐ Its helps to discharge sperm.
 - ☐ Produce and secrete male sex hormones.
- The primary hormones that help the male reproductive system function include:
 - ☐ **Follicle-stimulating hormone (FSH):** Your pituitary gland makes FSH. FSH is necessary to produce sperm (spermatogenesis).
 - ☐ **Luteinizing hormone (LH):** Your pituitary gland also makes LH. LH is necessary to continue the process of spermatogenesis.
 - ☐ **Testosterone.** Testosterone is the main sex hormone in males. It helps you develop certain characteristics, including muscle mass and strength, fat distribution, bone mass.
- Male sex organs can be divided into primary and secondary sex organs
- **Primary Sex Organs:**
 - ⇒ **Testes** are the primary sex organs in males.
 - ⇒ These are directly involved in the production of sperm and hormones.
 - ⇒ **Testes:**
 - ☐ There are two testicles in Male body.
 - ☐ Sperm are made in the testes.
 - ☐ Scrotums are the pouch like structures that carries testes and it is kept outside the body because sperm need a lower temperature to form properly and temperature inside the body is relatively higher.
 - ☐ The **testes** also release the hormone **testosterone**, which helps in sperm production.
 - ☐ Testosterone causes physical changes in boys during puberty, like a deeper voice and more body hair. These changes reflect the male characteristics.

⇒ **Sperm:**

- ❑ Sperm or spermatozoa, are male gametes that are produced in the testes and are designed to fertilize an egg.
- ❑ A mature sperm cell is about 2-5 micrometres long.
- ❑ Sperm have three main parts: the head, the middle piece, and the tail:
- ❑ Head of sperm contains the nucleus, which contains DNA that is tightly packed. The tip of the head is called the acrosome, which contains enzymes that help the sperm penetrate the egg.
- ❑ Middle piece of sperm contains mitochondria that provide energy for the sperm's movement.
- ❑ Tail of sperm also known as a flagellum is a long structure that enables sperm to swim.
- ❑ The tip of the sperm head of sperm is called as acrosome, which enables the sperm to penetrate the egg.

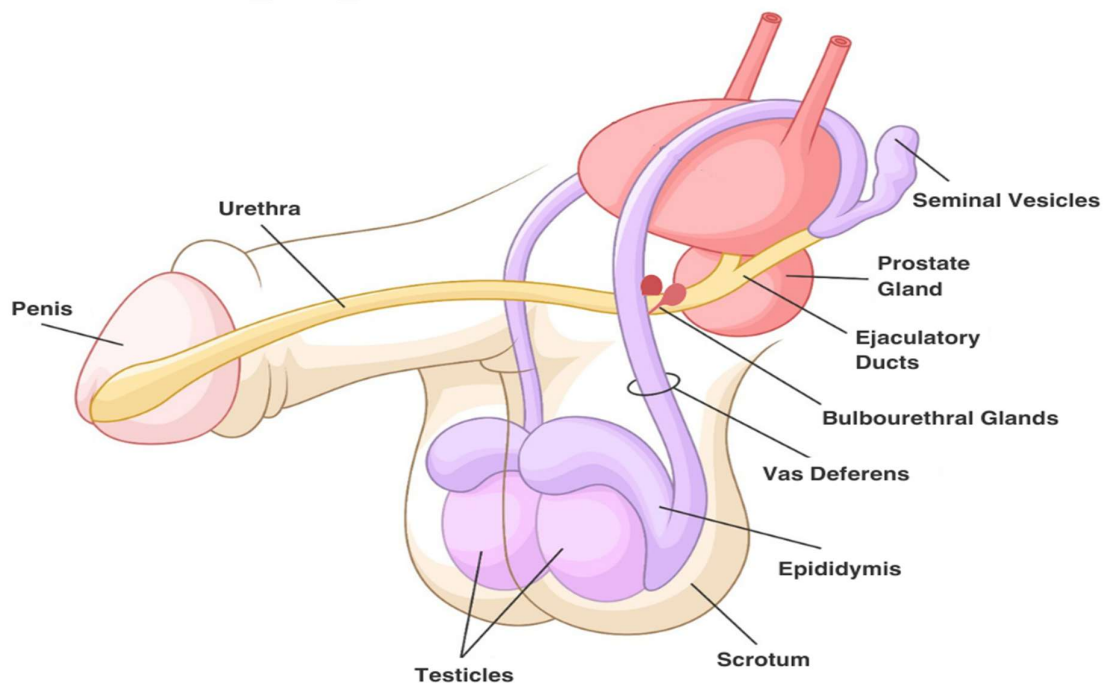


● Secondary Sex Organs:

- ⇒ These develop during puberty and help in the process of reproduction, but are not directly involved in sperm production.
- ⇒ Various Secondary sex organs are:

(1) Epididymis

- ❑ Epididymis is a long, coiled tube that rests on the back of each testicle.
- ❑ It carries and stores the sperm cells that your testicles create. Epididymis also brings the sperm to maturity.



(2) Vas deferens:

- ❑ Vas deferens is a long, muscular tube that travels from the epididymis into pelvic cavity, just behind urinary bladder.
- ❑ Vas deferens transports mature sperm to the urethra in preparation for ejaculation.
- ❑ They are also called as sperm ducts.

(3) Prostate gland:

- ❑ The prostate lies just under the bladder and surrounds the urethra.
- ❑ It is Walnut-sized in young men, the enlarges with age.
- ❑ The **prostate gland** has several important functions:
 - ⊙ **Produces Prostatic Fluid:** The prostate makes a fluid that makes up 20-30% of semen. This fluid is alkaline, helping sperm survive longer in the acidic vaginal tract, and also nourishes and protects sperm.
 - ⊙ **Helps with Ejaculation:** The prostate contracts during ejaculation to push semen through the urethra, mixing its fluid with sperm.
 - ⊙ **Supports Urinary Control:** The prostate surrounds the urethra and helps control urine flow, preventing backflow and aiding in bladder control.
 - ⊙ **Protects Against Infections:** Prostatic fluid contains substances that help fight infections in the urinary and reproductive systems.

(4) Seminal vesicles:

- ❑ Seminal vesicles are sac-like pouches that attach to your vas deferens near the base of your bladder.
- ❑ Seminal vesicles make up to 80% of your ejaculatory fluid.
- ❑ Functions of the fluid of Seminal Vesicle:
 - ⊙ It has Fructose, which provides energy.
 - ⊙ It has Prostaglandins, which suppresses the female immune response.
 - ⊙ Alkaline fluid, which helps keep the sperm alive in the acidic medium of vagina.

(5) Bulbourethral (Cowper) glands

- ❑ The bulbourethral glands are pea-sized and located on the sides of the urethra, just below the prostate.
- ❑ These glands produce a clear, slippery fluid.
- ❑ Functions of the fluid of Cowper Gland:
 - ⊙ The fluid lubricates the urethra, making it easier for sperm to travel.
 - ⊙ It neutralizes any remaining acids from urine, making the environment safer for sperm.

(6) Urethra:

- ❑ It is a tube that extends from the bladder to the external opening at the end of the penis.
- ❑ The urethra carries both urine and sperm.

(7) Penis

- ❑ Penis used for urination and sexual intercourse.
- ❑ It contains the urethra, which carries both urine and semen.
- ❑ It has spongy tissue that can fill with blood to cause an erection.

Female Reproductive System:

- The female reproductive system consists of organs and glands that allow to reproduce and have sexual intercourse.
- **Functions of Female Reproductive System:**
 - ❑ **Egg Production:** The ovaries produce female gametes or eggs needed for reproduction.
 - ❑ **Hormone Production:** The ovaries release hormones like oestrogen and progesterone, which regulate the menstrual cycle and secondary sexual characters.

- ❑ **Menstrual Cycle:** Regulates the monthly cycle, preparing the uterus for pregnancy and shedding the lining if pregnancy does not occur.
- ❑ **Fertilization:** Helps the Eggs to get fertilized by the sperm.
- ❑ **Pregnancy Support:** The uterus nurtures the fertilized egg during pregnancy.
- ❑ **Labor and Birth:** It helps to deliver the baby through vagina.
- ❑ **Lactation:** Helps the Breasts produce milk after childbirth to feed the baby.

- Female sex organs can be divided into primary and secondary sex organs:

- **Primary Sex Organs in Female:**

⇒ **Ovaries** are primary sex organs in female.

⇒ **Structure and location of Ovary:**

- ❑ Most females have two ovaries, one on each side of the uterus.
- ❑ They are almond-shaped and about the size of an almond.

⇒ **Functions of Ovary:**

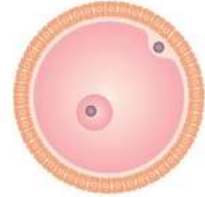
- ❑ The female germ-cells (eggs) are produced in the ovaries.
- ❑ The ovaries also produce hormones like Oestrogen and Progesterone which helps to regulate sexual characters.

⇒ **Egg Development by Ovary**

- ❑ At birth, a girl's ovaries already contain thousands of immature eggs called Oocyte.
- ❑ When a girl reaches puberty, some of these eggs start maturing, with one egg being produced each month by one of the ovaries.
- ❑ At puberty, these eggs begin to mature.
- ❑ Once a month, the ovaries release a mature egg, known as ovulation.

⇒ **Ovum:**

- ❑ An ovum or matured egg cell, is a large cell in the female reproductive system that is produced by Ovary.
- ❑ It is spherical in shape with a diameter of around 120 micrometres.



- **Secondary Sex Organs in Female:**

(1) **Oviducts (Fallopian Tubes):**

⇒ The fallopian tubes are narrow passageways that connect the ovaries to the uterus.

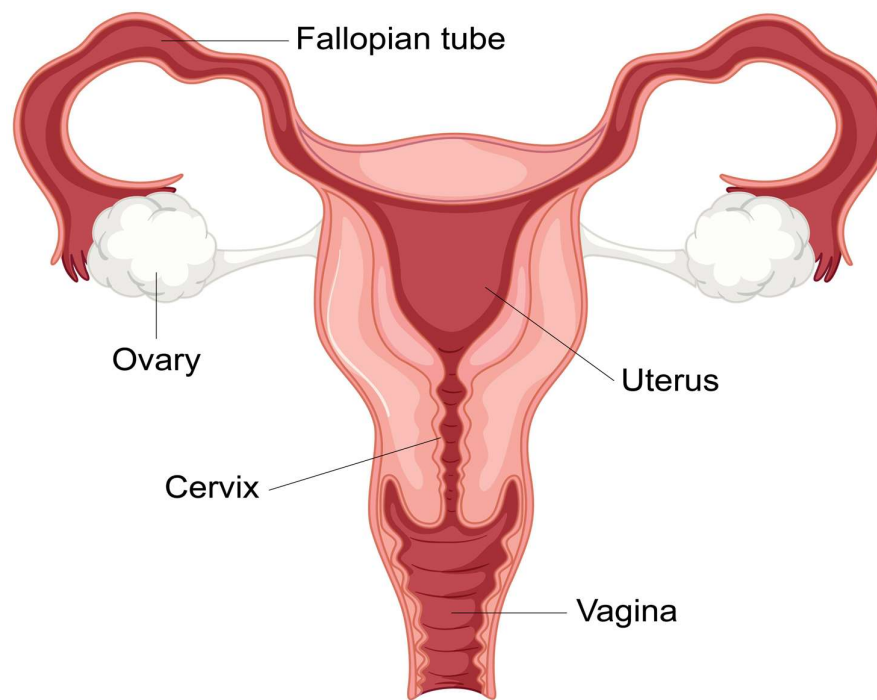
⇒ These are also called as Oviducts.

⇒ **Structure and location of Fallopian Tube:**

- ❑ They have funnel-shaped opening near the ovaries.
- ❑ They have finger-like projections that help catch the egg.
- ❑ They have tiny hair-like structures called cilia that move the egg toward the uterus.
- ❑ There are two fallopian tubes, one on each side, and they join together at the uterus.

(2) **Functions of Oviducts/Fallopian Tubes:**

- ❑ The fallopian tubes carry eggs from the ovaries to the uterus.
- ❑ They are the site where fertilization usually occurs.
- ❑ The fallopian tubes are narrow passageways that connect the ovaries to the uterus.



(3) Uterus:

⇒ The uterus is a hollow, pear-shaped organ in the female pelvis where a fertilized egg gets implanted and develops into a foetus during pregnancy.

⇒ It is also called the womb.

⇒ **Structure and location of Uterus:**

- ❑ The uterus is an elastic, bag-like structure.
- ❑ It has muscular walls and a lining called the Endometrium.
- ❑ Uterus is divided into two parts: the cervix and the corpus.
- ❑ Corpus is the upper and larger part of uterus that expands during pregnancy and Cervix is the lower part.
- ❑ Each month, the uterus prepares itself to receive a fertilized egg.
- ❑ The lining of the uterus (Endometrium) thickens and is richly supplied with blood to provide nourishment for the growing embryo.
- ❑ The uterus opens into the vagina through the cervix.

⇒ **Function of Uterus:**

- ❑ **Menstruation:** The uterus sheds its lining (endometrium) as blood during a period if pregnancy does not occur.
- ❑ **Pregnancy:** If an egg is fertilized, it implants in the uterine lining and begins to grow into an embryo and later a foetus.
- ❑ **Childbirth:** The uterus contracts to help push the baby out when it's time for birth.

(4) Cervix:

⇒ The cervix is the lower portion of the uterus of the female reproductive tract.

⇒ **Structure and location of Cervix:**

- ❑ The cervix is a narrow structure at the bottom of the uterus.

⇒ **Function of Cervix:**

- ❑ The cervix makes mucus that prevents sperm from entering when not fertile or during pregnancy.
- ❑ The mucus made by cervix also stops bacteria from getting into the uterus, helping keep the vagina healthy.

(5) Vagina:

⇒ The vagina is a muscular, elastic tube that's a vital part of the female reproductive system and sexual health.

⇒ Structure and location of Vagina:

- ❑ It sits behind the bladder and in front of the digestive tract.
- ❑ The vagina is a flexible tube connecting cervix to an external female reproductive organ.
- ❑ It is lined with mucous membranes that keep it moist.

⇒ Function of Vagina:

- ❑ The vagina can widen to allow a baby to pass during delivery and then shrink back to accommodate something narrow, like a tampon.
- ❑ The vagina lets menstrual blood and discharge leave the body.
It also allows sperm to enter the body during natural mating or artificial insemination.

+ Fertilization:

⇒ Fertilization is the process where a sperm and egg fuse to form a zygote, which then develops into a new organism.

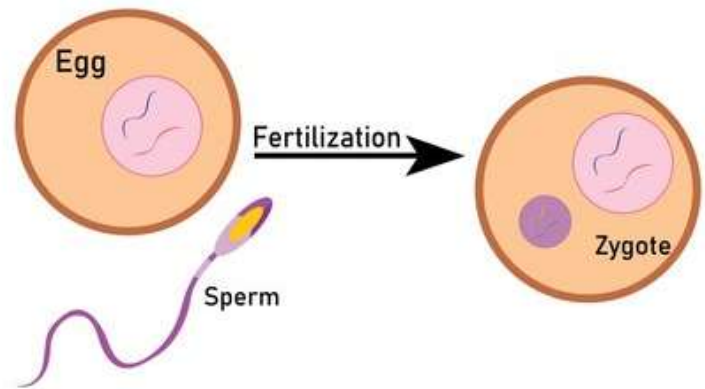
⇒ It takes place in the fallopian tube.

⇒ During sexual intercourse, sperms enter the vagina.

⇒ The sperm travel upwards through the cervix and the uterus to the fallopian tubes, where they may meet and fertilize the egg.

⇒ It results in the development of a zygote in the fallopian tube.

⇒ The fertilized egg (zygote) then gets implanted in the lining of the uterus, after 6-8 days, where it starts dividing and growing.



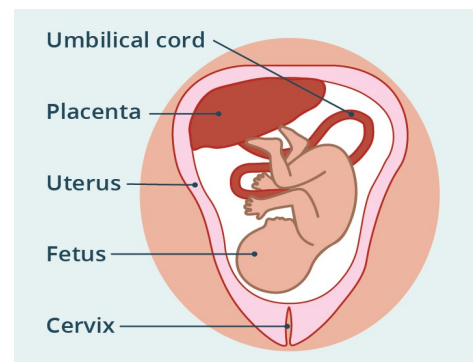
+ Embryo Implantation:

⇒ Embryo:

- ❑ Embryo is an early stage in the development of humans and other animals or plants.
- ❑ This stage lasts from shortly after fertilization until all major body parts appear.
- ❑ Major organs and structures, like the heart and brain, begin to develop.
- ❑ This stage lasts up to 8 weeks of pregnancy.

⇒ Placenta:

- ❑ The placenta is a special tissue embedded in the uterine wall.
- ❑ The placenta is a disc-shaped structure embedded in the uterine wall.
- ❑ The placenta has villi (small finger-like projections) on the embryo's side, and blood spaces on the mother's side surrounding these villi.
- ❑ The placenta is connected to a developing baby by a tube called the **umbilical cord**.
- ❑ Via the umbilical cord, nutrients like glucose and oxygen pass from the mother to the embryo, while waste products (like carbon dioxide) are transferred from the embryo into the mother's blood for removal.



⇒ **Implantation:**

- ❑ Implantation is the process in which the fertilized egg, or embryo, is attached to the uterus lining.
- ❑ This connection ensures the exchange of nutrients and waste between the mother and the embryo.
- ❑ Implantation can be done by development of placenta.

Pregnancy:

- ⇒ Pregnancy is the time during which offspring develops inside a woman's uterus.
- ⇒ It is also called as **Gestation period**.
- ⇒ It usually lasting around 40 weeks or just over 9 months as measured from the last menstrual period to delivery.
- ⇒ Stages of Offspring Development during Pregnancy:
 - ❑ **Zygote:** The sperm and egg combine to form a zygote. This is the first stage of development.
 - ❑ **Embryo (Up to 8 Weeks):** The zygote implants in the uterus and forms an embryo. Major organs and structures, like the heart and brain, begin to develop.
 - ❑ **Foetus (After 8 Weeks):** After 8 weeks, the embryo becomes a foetus. The foetus continues to grow, and its organs become more developed and functional.
 - ❑ **Foetal Growth and Development (Weeks 9-40):** The foetus grows rapidly, with all organs maturing. By the end of pregnancy, the foetus is fully developed and ready for birth.

Baby Birth and Breast-Feeding:

⇒ **Baby Birth:**

- ❑ The birth process is initiated by rhythmic contractions of the muscles in the uterus, which help to push the baby out.
- ❑ The cervix begins to soften, thin, and open up to allow the baby to pass through the birth canal.
- ❑ Labor is the process of delivering the baby through Vagina.
- ❑ After delivery Umbilical cord is cut, and the baby starts breathing.

⇒ **Breast-Feeding:**

- ❑ Following birth, generally mothers begin breastfeeding to provide the baby with essential nutrients and antibodies.
- ❑ This also helps the mother's body produce oxytocin hormone that helps the uterus to recover.

Menstruation:

- ⇒ If the egg is not fertilized, it survives for about one day.
- ⇒ Each month, the ovary releases an egg, and the uterus prepares by thickening its lining (Endometrium) to receive a fertilized egg.
- ⇒ **Menstruation (Menstrual Cycle)** is the monthly process in female body in which the lining of the uterus sheds and is expelled from the body through the vagina.
- ⇒ It involves the release of blood and other materials and occurs when an egg is not fertilized during the menstrual cycle.
- ⇒ This process usually lasts 2 to 8 days and happens roughly every month.

Methods to keep Reproductive organs Healthy:

⇒ Standard of living is very important for proper reproductive health. An expanding population makes it harder to improve standards of living, but social inequality can also be a major reason for poor living standards, irrespective of population size.

⇒ Here are some ways to keep your reproductive system healthy:

(i) Eating Balanced diet:

- ☐ Eat a healthy diet with plenty of calcium, magnesium, vitamin D, and folic acid.
- ☐ Avoid foods high in cholesterol, saturated fat, trans fat, and processed foods.

(ii) Exercise: Try to get at least three days of physical activity each week, mixing cardio exercises with strength training.

(iii) Practice good genital hygiene: Clean the genital area with soap and water once every day

(iv) Avoid smoking, excessive drinking and consuming other harmful drugs

(v) Manage Stress:

- ☐ Stress can disrupt the proper communication between the ovaries and the brain, which can cause menstrual issues

 Factors affecting sexual Activity:

⇒ Sexual maturation happens gradually and during overall body growth.

⇒ Reaching some level of sexual maturity doesn't mean the body or mind is ready for sexual activity or to have and raise children.

⇒ The question of when someone is ready for these responsibilities is complex, and we need to consider both physical and mental readiness.

⇒ Different factors affecting sexual activities:

- ☐ **Peer Pressure:** Friends may encourage participation in certain activities, even if someone isn't interested.
- ☐ **Family Pressure:** Families might push for marriage and to grow family.
- ☐ **Government Pressure:** There may be advice or pressure from authorities to avoid having children or to reproduce more for children. This may be done to control population.

 Sexually Transmitted Diseases:

⇒ Sexually transmitted Diseases (STD) are infections you can develop after engaging in sexual activities.

⇒ It's important to consider the possible health risks of sexual activity. Many diseases can be transmitted through sexual contact.

⇒ **Bacterial Infections:** STD caused by bacterial infection are:

- (i) Gonorrhoea: A bacterial infection that can cause genital discharge and painful or burning urination.
- (ii) Syphilis: Syphilis is a infection that causes sores and a skin rash

⇒ **Viral Infections:** STD caused by Viral infection are:

- (i) Warts: Genital warts are flesh-coloured bumps that can appear in clusters or individually in the genital area.
- (ii) HIV/AIDS: AIDS (acquired immunodeficiency syndrome) is the final stage of HIV (human immunodeficiency virus) infection which can weaken the immune system and causes death.

⇒ It is possible to reduce the risk of transmitting such diseases during sex by using protection. A condom, when used on the penis, can help prevent the transmission of many STDs to some extent.

 Contraception:

- ⇒ Sexual act has high possibility of pregnancy, if a woman is not prepared for pregnancy, it can place major physical and emotional demands on her body, which may adversely affect her health.
- ⇒ **Contraception** is the intentional prevention of conception through the use of various devices, sexual practices, chemicals, drugs or surgical procedures.
- ⇒ To prevent pregnancy, several contraceptive methods are available, falling into different categories:
- **Mechanical Barriers:** Methods such as condoms worn on the penis or similar coverings worn in the vagina act as physical barriers to prevent sperm from reaching the egg.
 - **Hormonal Methods:** These contraceptives alter the body's hormonal balance, preventing the release of eggs and fertilization. They are commonly taken orally as pills, but can have side effects due to hormonal changes.
 - **Intrauterine Devices (IUDs):** Devices like the loop or copper-T are placed in the uterus to prevent pregnancy. Although effective, they may cause irritation of the uterus and other side effects.
 - **Surgical Methods:**
 - ❑ The methods are operative procedures which block the passage of semen in males and ova in females.
 - ❑ The techniques are also called sterilisation procedures.
 - ❑ These methods prevent fertilization but require careful consideration, as improper surgery may lead to infections and other complications.
 - ❑ There are two types of Surgical methods based on gender:
 - (i) **Vasectomy**
 - © A short procedure for men that involves removing a small part of the vas deferens, which prevents sperm from the epididymis to pass to the seminal vesicles.
 - © Vasectomies are nearly 100% effective.
 - (ii) **Tubectomy:**
 - © It is a surgical procedure that permanently blocks, clips, or removes a woman's fallopian tubes to prevent pregnancy.
 - © Tubectomy is also known as tubal ligation.

Female Sex Ratio & Illegal Sex Determination

- © Prenatal sex determination is prohibited by law because illegal sex-selective abortion of female foetuses has led to an alarming decline in the child sex ratio in many sections of society.